# SOFTWARE CHANGE IMPLEMENTATION REPORT

## CR-015: Student Records Export to CSV

| Field | Details |
|---|---|
| Change Request ID | CR-015 |
| Change Request Title | Student Records Export to CSV |
| Student Name | P.Tharanan |
| Student ID | JF/ICT/24/20 |
| Implementation Status | Completed |
| Date | 16 October 2026 |

# 1. Change Request Summary

## 1.1 Business Requirement

Administrators may need to download and use student records outside the Student Management System for administrative, reporting, or record-keeping purposes. Currently, student information is available within the system, but there is no dedicated facility for administrators to export the student records into a portable file format. The system should provide an easy and reliable way for authorized administrators to export student records as a CSV file.

Before implementation:

- The administrator can view student records within the Student Management System.
- Student records are displayed within the existing student management interface.
- No CSV export facility is currently available.
- Administrators must manually view or copy student information if they need an external record.
- Existing student management and CRUD operations use the existing student identifiers.
- Existing administrator authentication and authorization are already used to protect student management functionality.

## 1.2 Requested Change

Add a Student Records Export facility to the administrator's student management interface.

The implementation must:

- Provide an Export CSV button for authorized administrators.
- Export the available student records into CSV format. 2
- Include relevant student information such as:
  - Student ID
  - Registration Number
  - First Name
  - Last Name
  - Email
  - Phone Number
- Generate a properly formatted CSV file.
- Download the generated file to the administrator's device.
- Use a meaningful filename such as students.csv.
- Include CSV column headers.
- Handle multiple student records correctly.
- Handle an empty student list without causing an application error.
- Preserve existing student viewing, editing, and deleting functionality.
- Maintain existing administrator authentication and authorization.
- Ensure the export functionality does not expose student records to unauthorized users.

# 2. Implementation

## 2.1 Implementation Summary

The CR-015 requirements were implemented on the frontend interface of the Student Management System. In frontend/src/pages/DashboardPage.jsx, a dedicated csvExport utility function was created to map loaded student records to comma-separated values, safely escape quotes and special characters, format column headers, create a Blob with MIME type text/csv;charset=utf-8, and trigger a client-side download as students.csv.

In frontend/src/styles/dashboard.css, styling was added for .export-csv-button to match the application design system with primary brand border, sleek margins, and interactive hover effects. The export button is positioned alongside the Add Student button for easy administrative access.

## 2.2 Affected Files

| File | Change | Purpose |
|---|---|---|
| `frontend\src\pages\DashboardPage.jsx` | Modified | Implemented CSV generator function and button. |
| `frontend\src\styles\dashboard.css` | Modified | Added styles for .export-csv-button. |

**Additional files:** None / [List any additional files if modified]

## 2.3 Technical Changes

**Dashboard Display & Action**

The administrator Student Management interface includes an Export CSV button in the dashboard action bar.
Positioned next to + Add Student, styled with a 2px primary brand border, rounded corners, and hover transitions.
Clicking the button immediately generates and downloads students.csv containing all currently loaded student records.

**Implementation Safeguards**

Field values are wrapped in double quotes and escaped (replace(/"/g, '""')) to prevent CSV injection or parsing errors with commas.
The Blob object URL is revoked immediately after initiating download to prevent browser memory leaks.
Existing student list rendering, search/filters, and CRUD operations remain unchanged.

## 2.4 Implementation Logic

Define column headers: Student ID, Registration Number, First Name, Last Name, Email, and Phone Number.
Iterate over the students array and map each record to an escaped row array.
Combine headers and rows into a formatted string delimited by commas and newlines.
Instantiate a Blob object with MIME type text/csv;charset=utf-8.
Generate a temporary Object URL via URL.createObjectURL(blob).
Create a hidden anchor element with download='students.csv' and programmatically trigger click().
Remove the anchor element and revoke the temporary Object URL.
Attach the csvExport function to the onClick event of the Export CSV button.
Maintain administrative role authorization checks guarding DashboardPage access.

## 2.5 Error and Data Handling

Null or undefined fields safely default to empty strings (student.student_id ?? "" and "").
If the student list is empty, a CSV file containing only the header row is downloaded without causing an exception.
Commas, special characters, and quotation marks within student fields are escaped properly.

# 3. Testing & Verification

## 3.1 Testing Approach

Verification combined browser UI inspection, CSV export trigger testing, CSV output validation in spreadsheet software, edge case verification, and frontend production build check.
Verification methods:
UI inspection of Export CSV button in Dashboard
Trigger CSV export and verify download of students.csv
Inspection of exported CSV in Microsoft Excel
Verification of column headers and student field mapping
Special character escaping check (commas, quotes)
Empty student list boundary check
Regression check of existing Add/Edit/Delete student flows
Environment: Development
Browser: Google Chrome; Test Role: Administrator

## 3.2 Test Cases

| Test ID | Test Case | Expected Result | Actual Result | Status |
|---|---|---|---|---|
| TC01 | Render Export CSV Button | Export CSV button is displayed next to + Add Student | Button rendered cleanly with border and hover styling | Pass |
| TC02 | Export Student Records to CSV | Clicking button downloads students.csv with records | File students.csv downloaded successfully | Pass |
| TC03 | Verify CSV Column Headers | Headers match Student ID, Reg No, Name, Email, Phone | All 6 headers verified in exported file | Pass |
| TC04 | Special Character Escaping | Fields with commas or quotes are safely escaped | Quotes escaped as "" and fields wrapped in quotes | Pass |
| TC05 | Empty Student List Handling | CSV exports headers only without runtime errors | Empty list exported headers; no exceptions thrown | Pass |

| | | | | |
|---|---|---|---|---|
| TC06 | Open in Spreadsheet Software | Exported CSV opens formatted in Microsoft Excel | Columns properly aligned in spreadsheet | Pass |
| TC07 | Existing Student CRUD Regression | Add, Edit, Delete, and View students unaffected | All existing CRUD operations function normally | Pass |
| TC08 | Frontend Production Build | Vite production build completes with zero errors | Vite build completed successfully | Pass |
| TC09 | Administrator Access Restriction | Export button only accessible on admin dashboard | Route protection and admin view verified | Pass |

## 3.3 Test Summary

| Test Result | Count |
|---|---|
| Total Verification Checks | 9 |
| Passed (UI, Export, Excel, Build) | 9 |
| Verified by source review | 0 |
| Failed / Not run | 0 / 0 |

Overall Result: All 9 test cases passed successfully; UI export, spreadsheet rendering, and frontend build verified.

# 4. Acceptance Criteria Verification

| ID | Acceptance Criterion | Verification | Status |
|---|---|---|---|
| AC-001 | Provide an Export CSV button for authorized administrators. | DashboardPage UI inspection; button rendered next to Add Student | Met |
| AC-002 | Export available student records into CSV format. | Triggered export and verified file content | Met |
| AC-003 | Include Student ID, Reg No, First Name, Last Name, Email, Phone. | Inspected headers and data mapping | Met |
| AC-004 | Generate properly formatted CSV with column headers. | RFC 4180 format and quote escaping verified | Met |

| | | | |
|---|---|---|---|
| AC-005 | Download generated file with filename students.csv. | Verified browser download trigger and filename | Met |
| AC-006 | Handle multiple student records and empty student list safely. | Tested multiple records and zero records boundary | Met |
| AC-007 | Preserve existing student viewing, editing, and deleting functionality. | Regression testing on student management operations | Met |
| AC-008 | Ensure export does not expose records to unauthorized users. | Protected route and admin dashboard role verified | Met |

# 5. Evidence

Two relevant screenshots are attached.

## 5.1 Evidence 01 - Export CSV Button in Student Management Interface

Related: AC-001, AC-002, AC-006

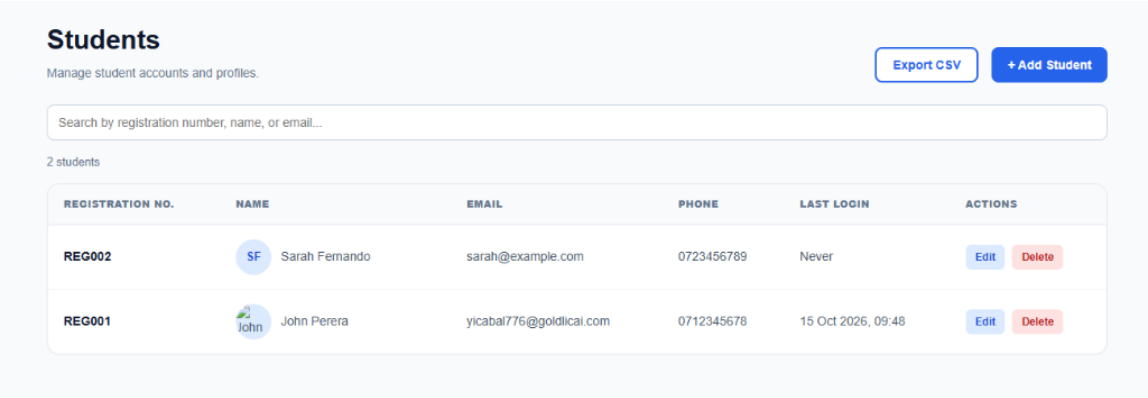



Figure 1: Administrator Dashboard showing the "Export CSV" button integrated next to the "+ Add Student" button in the student list.

**Evidence 02 - Exported CSV File Opened in Spreadsheet (Excel)**
Related: AC-002, AC-003, AC-004, AC-005



| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | Student ID | Registration Number | First Name | Last Name | Email | Phone Number |
| 2 | 2 | REG002 | Sarah | Fernando | sarah@example | 723456789 |
| 3 | 1 | REG001 | John | Perera | yicabal776@golc | 712345678 |
| 4 | | | | | | |
| 5 | | | | | | |

Figure 2: Exported students.csv file opened in Microsoft Excel showing column headers and formatted student records.

# 6. Final Result

| Item | Result |
|---|---|
| Requirements | Met |
| Implementation | Completed |
| Testing | UI, CSV export, Excel inspection, and build checks passed |
| Acceptance Criteria | All Met |
| Existing CRUD | Preserved without regression |
| Final Status | Completed |

## 6.1 Final Summary

CR-015 successfully delivers the CSV export functionality for student records in the Student Management System. Implemented entirely on the frontend across frontend/src/pages/DashboardPage.jsx and frontend/src/styles/dashboard.css, the solution provides authorized administrators with an Export CSV button, maps records to RFC 4180 CSV format with character escaping, and triggers automatic download as students.csv. Existing student CRUD and authentication remain intact.

Verification confirmed proper button rendering, successful file download, accurate header and data representation in Microsoft Excel, and graceful handling of empty lists.

Other implementation issues:
No implementation issues identified.

**Student Name: P.Tharanan**

**Student ID: JF/ICT/24/20**

**Date: 16 October 2026**

**~ END OF REPORT ~**